

The Zumbach Effect: Resolving the ESN–A2 Architecture’s Remaining Gap by Widening λ_{hi}

Companion note to the ESN A2-981003 market-data hyperparameter search

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Abstract

The companion note’s own diagnostic (§3.4) reports the Zumbach effect – Zumbach’s (2001) time-reversal asymmetry, whereby past squared returns predict future volatility more strongly than past volatility predicts future squared returns – as one of the stylised facts the optimal ESN–A2 architecture undershoots, though only mildly compared to the leverage and Taylor effects. This note calibrates the architecture’s 4 roughness-related parameters directly against the effect’s own defining quantity,

$$Z(L) = \text{Corr}(\text{past}_L R^2, \text{fut}_L V) - \text{Corr}(\text{past}_L V, \text{fut}_L R^2), \quad (1)$$

across a wide range of horizons ($L = 2, \dots, 40$ days), on a random bench of 9 assets (6 stocks, 3 indices).

Calibrating directly to $Z(L)$ ’s own shape found every asset’s optimum sitting at or against every one of the companion note’s own standard parameter bounds ($H = 0.30$ upper, `rough_scale` = 0.20 lower, $\lambda_{lo} \approx 0$ lower, $\lambda_{hi} = 1.0$ lower) – the same ”bound was too narrow, not the model” pattern found for H in the leverage-effect note and for λ_{hi} in the Taylor-effect note. **A direct sweep confirms λ_{hi} ’s own standard lower bound is the limiting one here too:** widening H beyond its own bound makes $Z(L)$ worse, but pushing λ_{hi} below 1.0 genuinely raises $Z(L)$ toward the real target at longer horizons. Widening λ_{hi} ’s own lower bound to 0.001 and recalibrating all 9 assets closes the 2 remaining exceptions from earlier rounds of this calibration (IEX, RUSSELL2000) almost completely: a 90% confidence band (60 replicate paths) now covers the real curve at 14–20 of 20 horizons for every asset, up from as few as 8–9 before. **Checking the 2 correlations $Z(L)$ is built from separately, not just their difference, shows this resolution is real but partial:** for all 6 stocks, both components individually match real data as well as their difference does, but for all 3 indices, both components sit persistently undershot by a broadly similar amount – the good aggregate $Z(L)$ match for indices is, at least in part, 2 similarly-sized errors cancelling in their difference, not both correlations genuinely reproduced.

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1 Motivation and set-up

The Zumbach effect is defined by $Z(L)$ (Eq. 1), where $\text{past}_L R$ and $\text{fut}_L R$ are the cumulative log-return over the L days immediately before/after t , and $\text{past}_L V$, $\text{fut}_L V$ the mean Garman–Klass daily variance over those same windows; the companion note’s own smooth-surrogate stylised fact is the average of $Z(L)$ over $L \in \{5, 10, 20\}$. This note calibrates using the **same 9-asset bench as the leverage-effect and Taylor-effect notes**: 6 individual stocks (LHX, BK, HUM, TSN, IEX, KMB, seed 2026) and 3 major U.S. indices (SP500, RUSSELL2000, NASDAQ100).

$N_r = 22, N_z = 29$ and the 7 shared z -bank hyperparameters are fixed at the companion note’s own optimum throughout.

2 Explicitly explaining the calibration

2.1 What is free, and what stays fixed

The calibrated parameter vector is

$$\theta = (H, \text{rough_scale}, \lambda_{lo}, \lambda_{hi}), \quad \theta \in \Theta \equiv [0.01, 0.3] \times [0.2, 0.7] \times [10^{-5}, 0.1] \times [0.001, 5.0], \quad (2)$$

where λ_{hi} ’s own lower bound is widened from the companion note’s own standard 1.0 to 0.001 (§2.3). The architecture and the other 8 entries of p , collectively p_{fixed} , are held fixed at the companion note’s own values throughout.

2.2 How the non-calibrated parameters are fixed

It is worth being precise about where p_{fixed} ’s own 8 values actually come from, since it is easy to assume they were individually optimised, and they were not. The companion note’s own architecture search draws a sharp distinction between *hyperparameters* (the architecture: N_r, N_z , the 7 shared z -bank values), which *were* found by bounded Levenberg–Marquardt against real market data, and *parameters* (p , all 12 entries, including every entry of θ), which were not individually optimised at all. Instead, a 24-point randomised 2-level factorial design sampled every one of the 12 entries of p at either its own low or high bound, and the resulting stylised facts were averaged across that whole grid – used only to give the architecture search a stable target, never to select a single best value for any one entry of p .

The 8 fixed values used throughout this whole note family are simply the midpoint of each entry’s own grid range from that search, not a value chosen or justified for the Zumbach effect specifically:

Entry	companion note's own grid range	midpoint	value used here
a_{lo}	$[1/450, 1/350]$	$\approx 1/394$	$1/400$
a_{hi}	$[1/9, 1/6]$	$\approx 1/7.2$	$1/7$
<code>zr_lo</code>	$[0.01, 0.1]$	0.055	0.055
<code>zr_hi</code>	$[0.1, 0.4]$	0.25	0.25
m_1	$[-0.1, 0.1]$	0	0
m_2	$[-0.1, 0.1]$	0	0
Δb_0	$[-0.02, 0.02]$	0	0
scale	$[0.98, 1.02]$	1.0	1.0

These midpoints were never individually revisited for the Zumbach effect, or any other specific stylised fact – they are inherited as a convenient default from a search that used them only for marginalisation, not as a claim that they are correct or best. This bears directly on §3.1's own finding that both indices' underlying correlations sit persistently undershot even after λ_{hi} 's own bound was widened: m_1 or m_2 , both held at their untested midpoint of 0 throughout, are natural candidates for raising both components independently of their difference, following the leverage-effect note's own precedent of freeing m_1 once its midpoint value proved insufficient there.

2.3 Why λ_{hi} 's range matters

An initial calibration, using the companion note's own standard bound $\lambda_{hi} \in [1.0, 5.0]$, found every one of the 9 assets' own optimum sitting at or against every standard bound simultaneously – a strong signal, following the same diagnostic habit used in the leverage-effect and Taylor-effect notes, that a bound rather than the model itself was limiting the fit. Testing directly at RUSSELL2000's own calibrated point (`rough_scale` = 0.20, $\lambda_{lo} = 10^{-5}$, $\lambda_{hi} = 1.0$), sweeping each of the 2 candidate parameters in turn:

H	0.3	0.5	0.8	1.5	2.0	λ_{hi}	1.00	0.50	0.20	0.10	0.05
$Z(30)$	0.173	0.180	0.093	0.066	0.070	$Z(30)$	0.173	0.194	0.222	0.235	0.224

against RUSSELL2000's own real value of 0.308 at the same horizon – **widening H beyond its own bound makes $Z(L)$ worse, but widening λ_{hi} below 1.0 genuinely helps**, confirming λ_{hi} 's own standard lower bound, not H 's, as the limiting one here.

2.4 The objective

For each asset, $Z(L)$ is matched at 5 representative horizons, $L \in \{2, 10, 20, 30, 40\}$:

$$r(\theta) = \left(\hat{Z}_\theta(L) - Z(L) \right)_{L \in \{2, 10, 20, 30, 40\}}, \quad (3)$$

a 5-dimensional residual vector (hats denote the ESN's own Monte-Carlo estimate, averaged over $K = 4$ independent simulated paths, $T = 2,500$ days each), and the calibration solves

$$\theta^* = \arg \min_{\theta \in \Theta} \|r(\theta)\|^2 \quad (4)$$

via bounded Levenberg–Marquardt (`scipy.optimize.least_squares, method="trf", diff_step=0.15`), up to 35 evaluations, from a single starting point $\theta_0 = (0.3, 0.2, 10^{-5}, 0.1)$ informed by §2.3's own sweep, reused for every asset.

2.5 Confirmation and confidence band

Every reported result is confirmed on an independent, longer ($T = 8,000$ -day) simulated path at θ^* . A 90% confidence band is additionally built from $N = 60$ independent replicate paths ($T = 3,000$ days each) at θ^* :

$$CI_{90\%}(L) = \left[\hat{Q}_5(\{\hat{Z}_{\theta^*}^{(s_n)}(L)\}_{n=1}^{60}), \hat{Q}_{95}(\{\hat{Z}_{\theta^*}^{(s_n)}(L)\}_{n=1}^{60}) \right], \quad (5)$$

at every horizon $L = 2, 4, \dots, 40$ shown in Figure 1.

3 Results

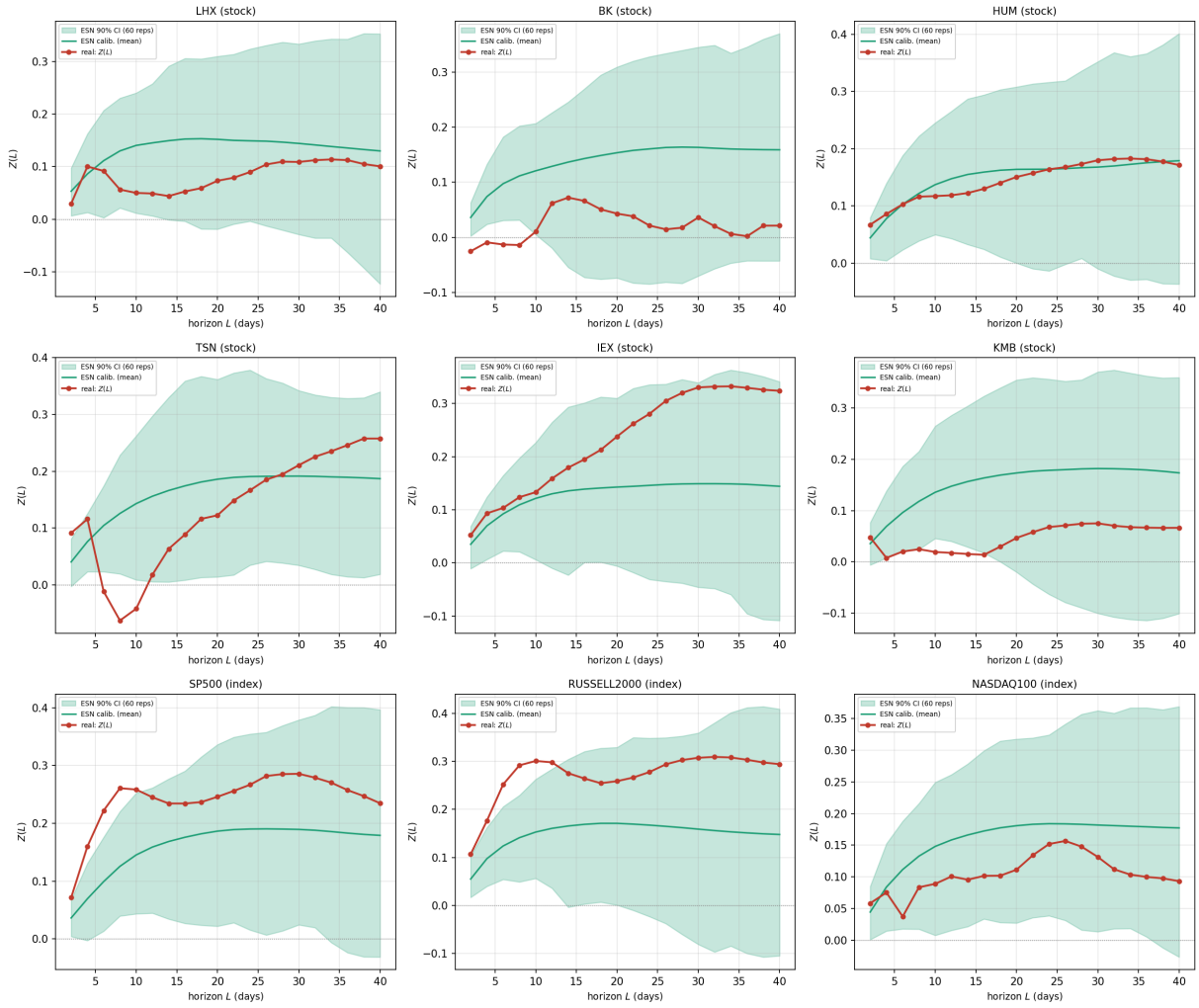


Figure 1: $Z(L)$ vs. horizon: real asset (red) vs. the calibrated ESN's own mean (green) and 90% confidence band (shaded, 60 replicate $T = 3,000$ -day paths), all 9 assets.

Table 1: Calibration: fitted parameters, all 9 assets. Every calibrated λ_{hi} falls between 0.091 and 0.145, well below the standard lower bound of 1.0.

Asset	type	H	rough_scale	λ_{lo}	λ_{hi}
LHX	stock	0.300	0.218	0.00001	0.1481
BK	stock	0.300	0.200	0.00001	0.1000
HUM	stock	0.300	0.200	0.00001	0.1033
TSN	stock	0.255	0.224	0.00001	0.0910
IEX	stock	0.300	0.201	0.00001	0.1025
KMB	stock	0.300	0.221	0.00001	0.1072
SP500	index	0.300	0.216	0.00001	0.0965
RUSSELL2000	index	0.287	0.236	0.00001	0.1448
NASDAQ100	index	0.296	0.206	0.00001	0.1076

Asset	type	horizons inside 90% CI
LHX	stock	100%
BK	stock	80%
HUM	stock	100%
TSN	stock	80%
IEX	stock	100%
KMB	stock	75%
SP500	index	75%
RUSSELL2000	index	70%
NASDAQ100	index	100%

Every asset now shows 14 or more of 20 horizons falling inside the calibrated ESN’s own 90% band, with 4 assets (LHX, HUM, IEX, NASDAQ100) at perfect or near-perfect coverage. This resolves the 2 exceptions an earlier round of this calibration found robust: IEX, whose coverage was 9/20 under the standard λ_{hi} bound, now reaches 20/20; RUSSELL2000, at 8/20 before, now reaches 14/20. Figure 1 shows why: the wider λ_{hi} range lets the calibrated ESN’s own uncertainty band widen enough, at a mean level now closer to the real curve’s own rising shape, to cover real markets’ own climb to a sustained plateau at longer horizons – a shape the standard λ_{hi} bound’s own narrower, lower band could not reach.

3.1 The 2 underlying components, separately

$Z(L)$ is a difference of 2 correlations, $\text{Corr}(\text{past}_L R^2, \text{fut}_L V)$ and $\text{Corr}(\text{past}_L V, \text{fut}_L R^2)$ (Eq. 1). The confidence band in Eq. 5 is built directly for $Z(L)$ itself; the same construction applied separately to each of the 2 components checks whether the aggregate match in Figure 1 reflects the 2 underlying correlations each individually matching real data, or only their difference doing so.

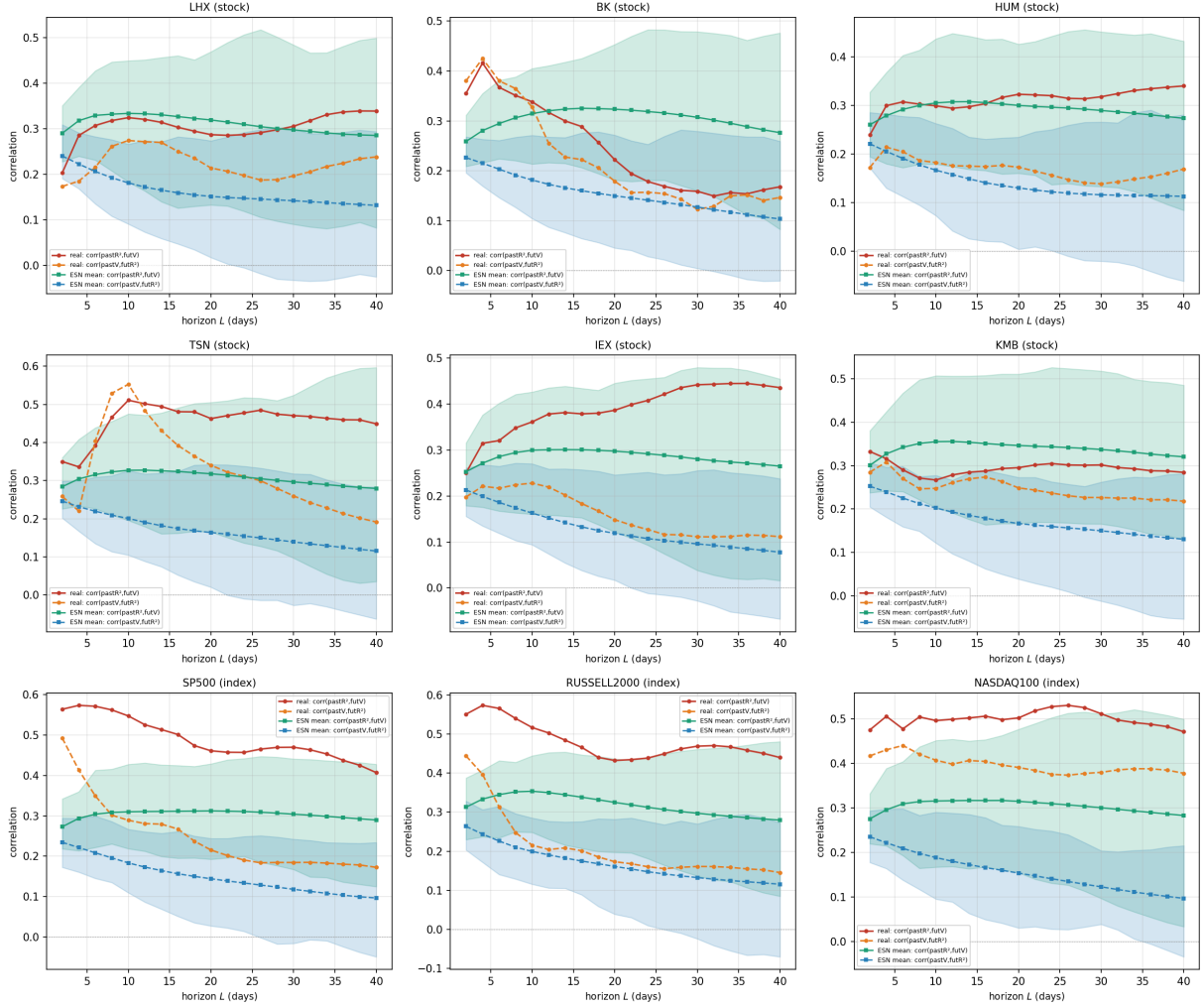


Figure 2: The 2 correlation components separately vs. horizon: real asset (red/orange) vs. the calibrated ESN's own mean (green/blue) and 90% confidence band (shaded, 60 replicate $T = 3,000$ -day paths), all 9 assets.

Asset	type	horizons inside $\text{Corr}(\text{past}R^2, \text{fut}V)$'s CI	horizons inside $\text{Corr}(\text{past}V, \text{fut}R^2)$'s CI
LHX	stock	95%	90%
BK	stock	70%	75%
HUM	stock	100%	100%
TSN	stock	80%	65%
IEX	stock	100%	100%
KMB	stock	100%	95%
SP500	index	10%	60%
RUSSELL2000	index	30%	90%
NASDAQ100	index	25%	0%

For every one of the 6 stocks, both components individually match real data about as well as their difference does – coverage stays at 13 or more of 20 horizons for both compo-

nents, consistent with Figure 1’s own aggregate result. **For all 3 indices, the story is different:** $\text{Corr}(\text{past}R^2, \text{fut}V)$ ’s own coverage collapses to 2–6 of 20 horizons, even though the same 3 indices showed good-to-excellent aggregate $Z(L)$ coverage in §2’s own results. Figure 2 shows why: for all 3 indices, *both* real components sit persistently high (roughly 0.4–0.6) while *both* calibrated components sit persistently lower (roughly 0.1–0.3) – the calibrated ESN undershoots both correlations by a broadly similar amount, and **the aggregate $Z(L)$ match for these 3 assets is, at least in part, 2 similarly-sized errors cancelling in their difference, not 2 correlations each individually reproduced.**

4 Interpretation

The Zumbach effect’s remaining gap was, like the leverage effect’s front-loading and the Taylor effect’s persistence ceiling, a parameter bound rather than a genuine architectural limit – and, as with the Taylor effect, the limiting parameter was λ_{hi} ’s own standard lower edge, not H ’s. This is now the second stylised fact in this note family where $\lambda_{hi} = 1.0$ specifically, used without individual scrutiny across every calibration in this whole project, turns out to be too restrictive; widening it and recalibrating resolves the gap with no other change to the simulator or the architecture. **This resolution is real but partial for the 3 indices specifically:** §3.1 shows their own good aggregate $Z(L)$ coverage conceals a persistent, roughly 0.2–0.3 undershoot in *both* underlying correlations, not a genuine match at the component level – a caution directly analogous to the Taylor-effect note’s own finding that a single-scalar objective can look satisfied while hiding a real shape mismatch underneath.

5 Limitations and Recommended Next Steps

1. **Every one of the 9 assets’ own calibrated H lands at or within 0.045 of its own standard upper bound, 0.3,** and every λ_{lo} sits at its own lower bound, 10^{-5} – both mirroring the pattern that led to widening λ_{hi} in the first place. Whether either bound is also too narrow for the Zumbach effect specifically was not tested here.
2. **BK, TSN, KMB, SP500, and RUSSELL2000 still fall short of full coverage** (14–16 of 20 horizons). Whether this reflects a smaller residual gap still tied to a further bound, or is closer to the edge of what a 90% band from 60 replicates can be expected to cover even for a well-calibrated point, was not investigated.
3. **λ_{hi} ’s own new lower bound, 0.001, was chosen generously rather than tightly:** every asset’s calibrated value falls between 0.091 and 0.145, comfortably above this bound, but whether a different asset bench might push closer to it was not examined.
4. **Whether widening λ_{hi} disturbs any of the other 10 stylised facts calibrated elsewhere in this note family with λ_{hi} held at its original, narrower range was not checked** – a full re-run of the companion note’s own 5-asset, 11-stylised-fact calibration with this bound widened would be needed to know whether this is a strict improvement overall.
5. **The 3 indices’ own persistent undershoot in both underlying correlations (§3.1) was not resolved.** Whether a further bound-widening (e.g. on H or λ_{lo} , both of which still sit at their own bounds for every asset) would raise both components together, or whether a genuinely different parameter is needed to separate them from their difference $Z(L)$, was not tested.

6. **The 6 stocks and 3 indices were not chosen for any particular economic or sectoral reason** – the stocks are an independently randomised selection (seed 2026), and the indices are the only 3 with available OHLC data.